

Multicommodity Flow Problem

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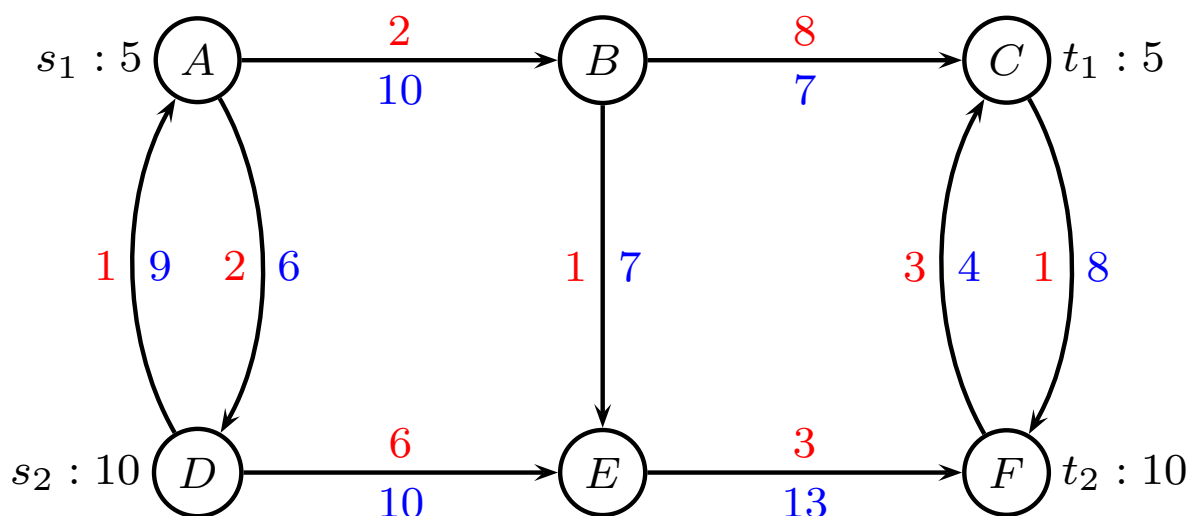
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Multicommodity Flow - Terminology

- We consider a directed graph $G = (V, E)$ in which each edge e has a capacity $u_e \in \mathbb{R}_+$ and a unit transportation cost $c_e \in \mathbb{R}$.
- A set K of commodities is given. Each commodity k has a source s_k , terminal t_k , and demand d_k .
- All commodities k should be transported from source to sink in the cheapest possible way, respecting capacities.



Applications

Direct network applications

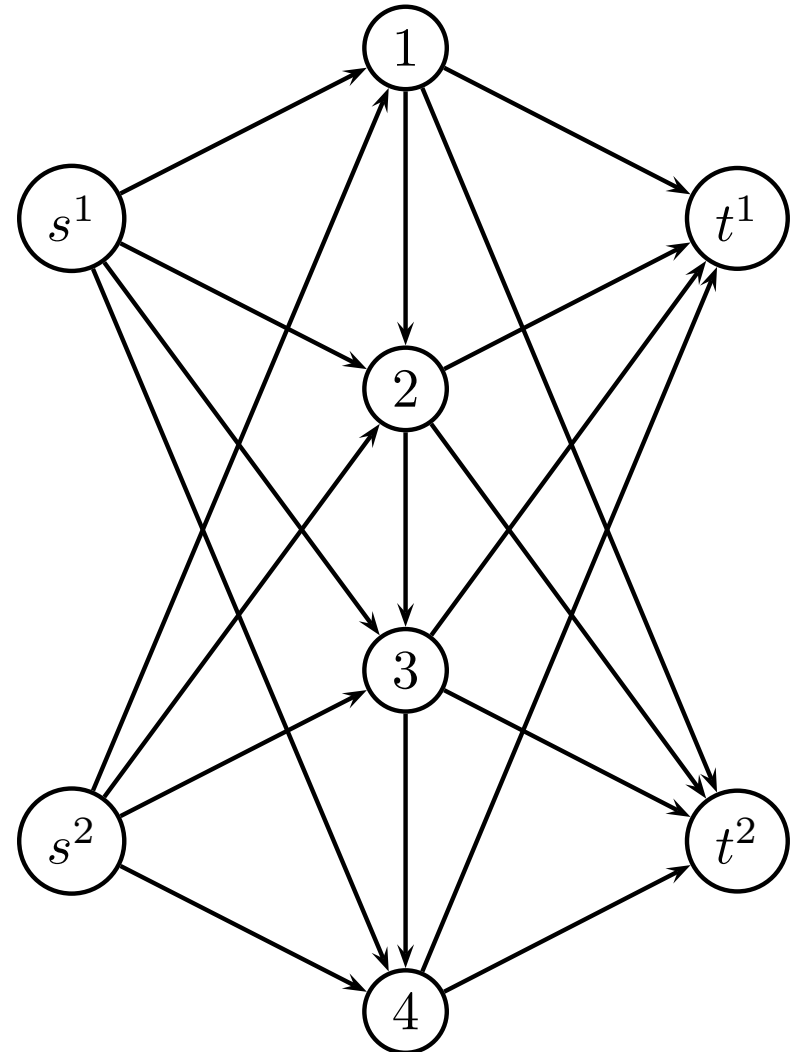
- Communication networks
- Computer networks
- Transportation network (busses, vehicles, trains)
- Foodgrain export-import network

Several commodities share capacity of network

Application: Warehousing

Seasonal production planning: four time periods i , two products k , shared inventory capacity u_i

- production cost p_i^k
- production capacity c_i^k
- storage cost h_i
- storage capacity u_i
- sales price z_i^k
- demand d_i^k



Multicommodity Flow, edge formulation

Let x_{ij}^k denote flow of commodity k on edge (i, j) .

$$\begin{aligned} \min \quad & \sum_{k \in K} \sum_{(i,j) \in E} c_{ij} x_{ij}^k \\ \text{s.t.} \quad & \sum_{i \in V} x_{ij}^k - \sum_{i \in V} x_{ji}^k = 0 & k \in K, j \in V \setminus \{s_k, t_k\} \\ & \sum_{i \in V} x_{ij}^k - \sum_{i \in V} x_{ji}^k = -d_k & k \in K, j = s_k \\ & \sum_{i \in V} x_{ij}^k - \sum_{i \in V} x_{ji}^k = d_k & k \in K, j = t_k \\ & \sum_{k \in K} x_{ij}^k \leq u_{ij} & (i, j) \in E \\ & 0 \leq x_{ij}^k \leq d_k & k \in K, (i, j) \in E \end{aligned}$$

Often integral flow: $x_{ij} \in \mathbb{Z}$

LP-problem hard to solve for large problems

IP-problem NP-hard to solve

MCF - Variants

- min total cost (normal formulation)
- variable cost for each k
- max flow (costs ignored)
- congestion
- splittable
- unsplittable
- k -splittable

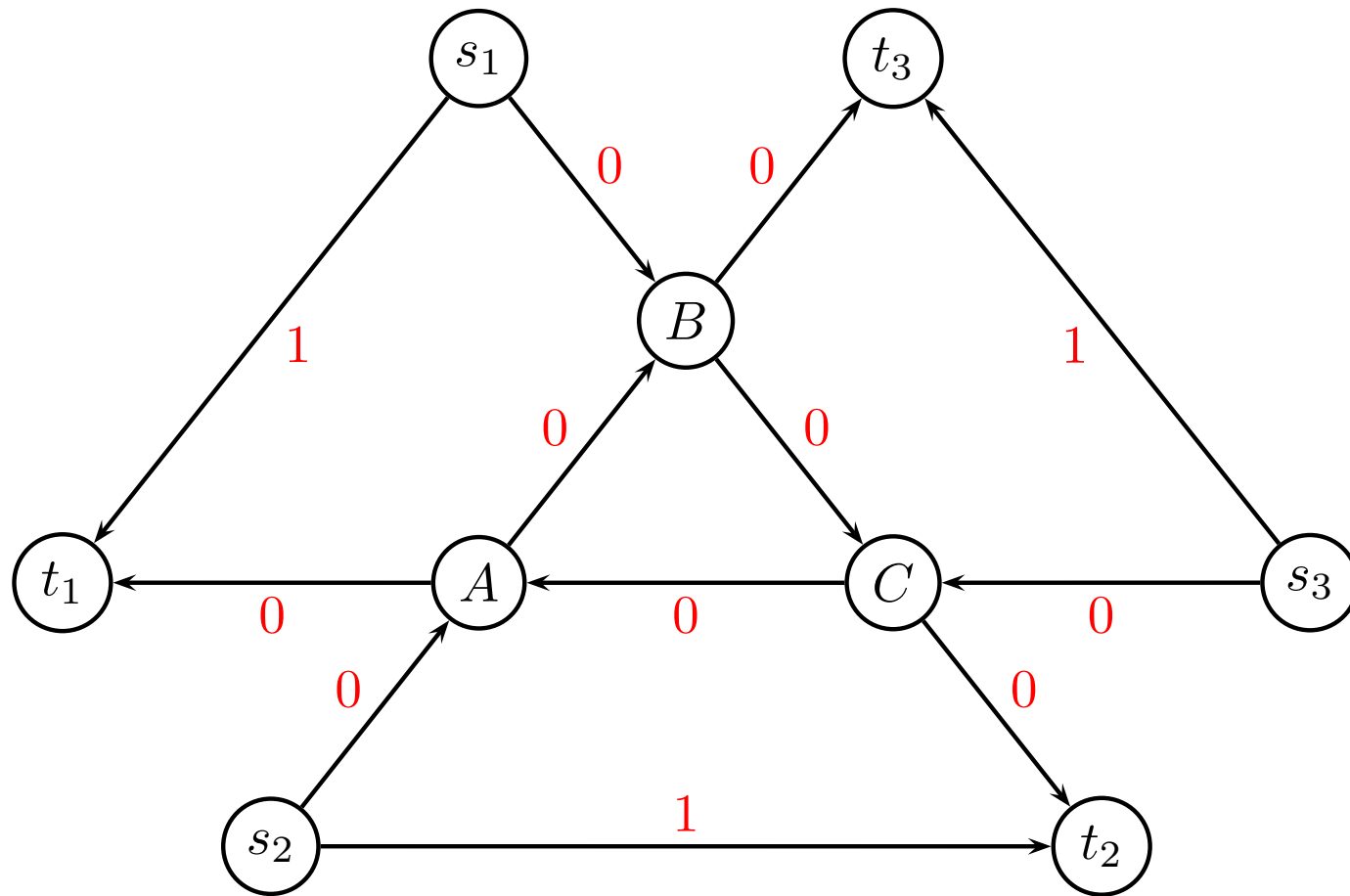
Congestion:

$$\min \sum_{(i,j) \in E} \frac{\sum_k x_{ij}}{u_{ij} - \sum_k x_{ij}}$$

Structure of LP model

	$x_{e_1}^1$	$x_{e_1}^2$	$x_{e_2}^1$	$x_{e_2}^2$	...	x_{ij}^k	...	$x_{e_m}^K$		
	$w_{e_1}^1$	$w_{e_1}^2$	$w_{e_2}^1$	$w_{e_2}^2$	...	w_{ij}^k	...	$w_{e_m}^K$		
1	-1	-1	1	-1	...	.	...	.	=	b_1
2	.	1	-1	.	...	1	...	.	=	b_2
.	.	.	.	.	...	.	...	.	=	.
.	.	.	.	.	...	.	...	.	=	.
n	.	.	.	1	...	-1	...	1	=	b_n
e_1	-1	.	-1	.	...	.	...	.	$\geq$	$-u_1$
e_2		-1	.	-1	...	-1	...	.	$\geq$	$-u_2$
.	.	.	.	.	...	.	...	.	$\geq$	.
e_m	.	.	.	.	...	.	...	-1	$\geq$	$-u_m$

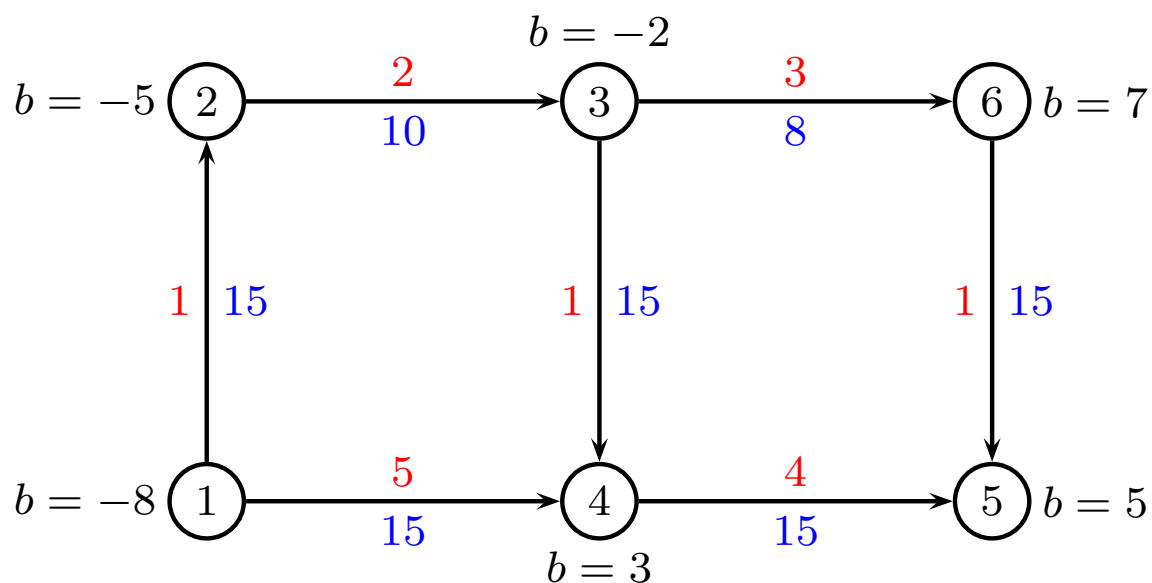
Fractional solutions



Demand $d_k = 1$ for all k . Capacity $u_e = 1$ for all e

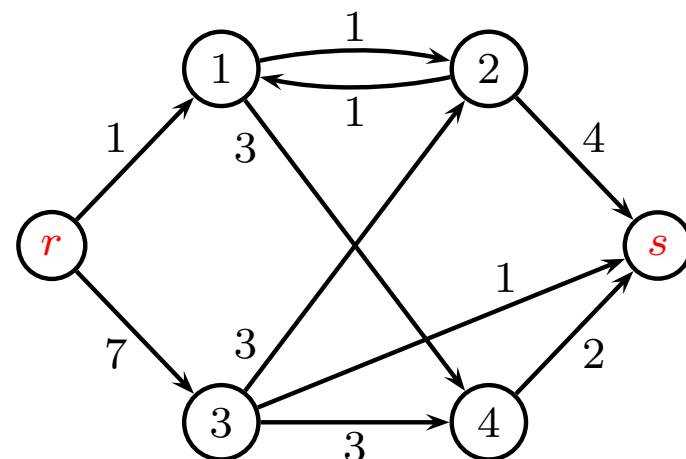
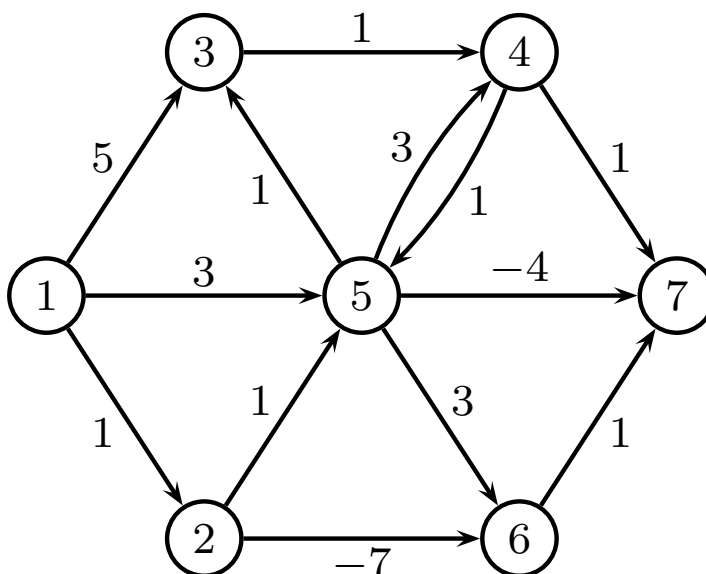
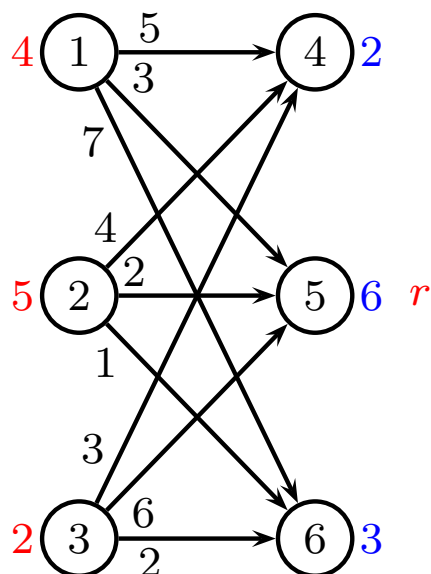
Special cases

MCF contains min cost flow as special case (one commodity)



Special cases

1. The Transportation Problem
2. The Shortest Path Problem
3. The Max Flow Problem



Solution methods

LP-based

- Simplex
- interior-point methods

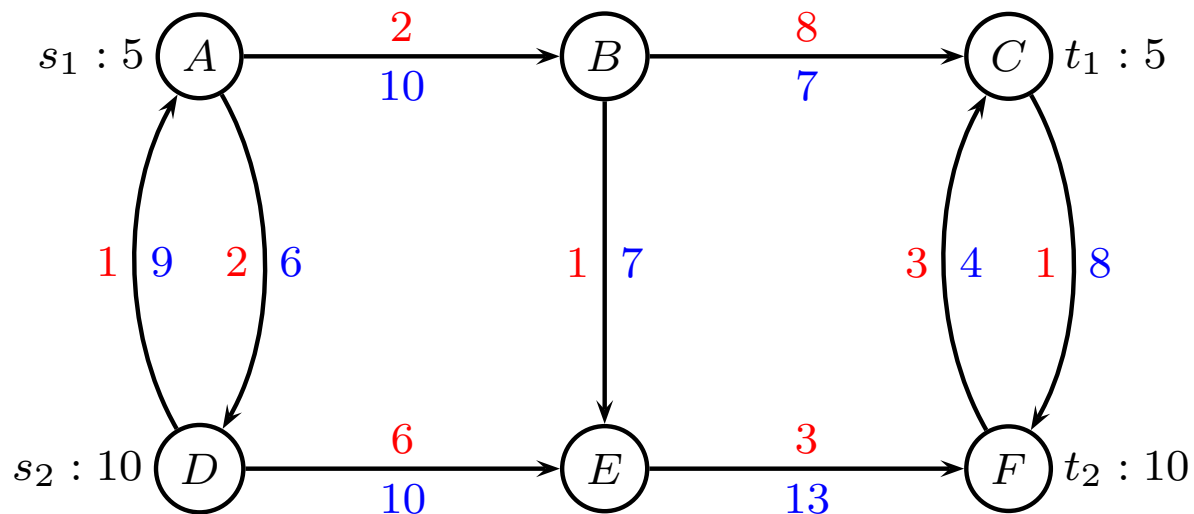
Column generation

- Dantzig Wolfe decomposition
- Master Problem
- Subproblem

Lagrangian methods

- Relax (delete) capacity constraints
- Punish costs if capacity violated

Multicommodity Flow - Example



All possible paths

path number	commodity k	path p	$\sum c_{ij}$
1	1	$A \rightarrow B \rightarrow C$	10
2	1	$A \rightarrow B \rightarrow E \rightarrow F \rightarrow C$	9
3	1	$A \rightarrow D \rightarrow E \rightarrow F \rightarrow C$	14
4	2	$D \rightarrow E \rightarrow F$	9
5	2	$D \rightarrow A \rightarrow B \rightarrow C \rightarrow F$	12
6	2	$D \rightarrow A \rightarrow B \rightarrow E \rightarrow F$	7

Path formulation

P_k set of possible paths from s_k to t_k

Binary $a_{p,ij}^k$ is 1 if path p for commodity k is using edge (i, j)

$\hat{c}_p^k$ denote the cost for path $p \in P_k$

x_p^k is flow on path p for commodity k

$$\begin{aligned} \min \quad & \sum_{k \in K} \sum_{p \in P_k} \hat{c}_p^k x_p^k \\ \text{s.t.} \quad & \sum_{k \in K} \sum_{p \in P_k} a_{p,ij}^k x_p^k \leq u_{ij} \quad (i, j) \in E \\ & \sum_{p \in P_k} x_p^k = d_k \quad k \in K \\ & 0 \leq x_p^k \leq d_k \quad k \in K, p \in P_k \end{aligned}$$

Dantzig-Wolfe reformulation

Generally easier to solve since fewer constraints

Path model

Set of paths: $P_1 = \{1, 2, 3\}$, $P_2 = \{4, 5, 6\}$.

minimize

$$10x_1 + 9x_2 + 14x_3 + 9x_4 + 12x_5 + 7x_6$$

subject to

$$\text{AB: } +x_1 + x_2 \quad \quad +x_5 + x_6 \leq 10 \quad (y_{\text{AB}})$$

$$\text{BC: } +x_1 \quad \quad +x_5 \leq 7 \quad (y_{\text{BC}})$$

$$\text{DE: } \quad \quad +x_3 + x_4 \leq 10 \quad (y_{\text{DE}})$$

$$\text{EF: } \quad +x_2 + x_3 + x_4 \quad +x_6 \leq 13 \quad (y_{\text{EF}})$$

$$\text{BE: } \quad +x_2 \quad \quad +x_6 \leq 7 \quad (y_{\text{BE}})$$

$$\text{AD: } \quad \quad +x_3 \leq 6 \quad (y_{\text{AD}})$$

$$\text{DA: } \quad \quad +x_5 + x_6 \leq 9 \quad (y_{\text{DA}})$$

$$\text{CF: } \quad \quad +x_5 \leq 8 \quad (y_{\text{CF}})$$

$$\text{FC: } \quad +x_2 + x_3 \leq 4 \quad (y_{\text{FC}})$$

$$\text{K1: } x_1 + x_2 + x_3 \geq 5 \quad (y^*_{\text{1}})$$

$$\text{K2: } x_4 + x_5 + x_6 \geq 10 \quad (y^*_{\text{2}})$$

Path model

P_k (paths from s_k to t_k) can be very large

Column generation (remember: fixed charge transportation problem)

- Master problem
- Restricted master problem $\overline{P}_k \subset P_k$
- Subproblem (generate a column — a path from s_k to t_k)

First iteration

Initial solution, paths 1 and 4. Restricted master problem:

minimize

$$10x_1 + 9x_4$$

subject to

$$\text{AB: } +x_1 \leq 10 \quad (y_{\text{AB}})$$

$$\text{BC: } +x_1 \leq 7 \quad (y_{\text{BC}})$$

$$\text{DE: } +x_4 \leq 10 \quad (y_{\text{DE}})$$

$$\text{EF: } +x_4 \leq 13 \quad (y_{\text{EF}})$$

$$\text{BE: } \leq 7 \quad (y_{\text{BE}})$$

$$\text{AD: } \leq 6 \quad (y_{\text{AD}})$$

$$\text{DA: } \leq 9 \quad (y_{\text{DA}})$$

$$\text{CF: } \leq 8 \quad (y_{\text{CF}})$$

$$\text{FC: } \leq 4 \quad (y_{\text{FC}})$$

$$\text{K1: } x_1 \geq 5 \quad (y^*_1)$$

$$\text{K2: } x_4 \geq 10 \quad (y^*_2)$$

Solution: $x_1 = 5, x_4 = 10$

Dual: $y_1^* = 10, y_2^* = 9$

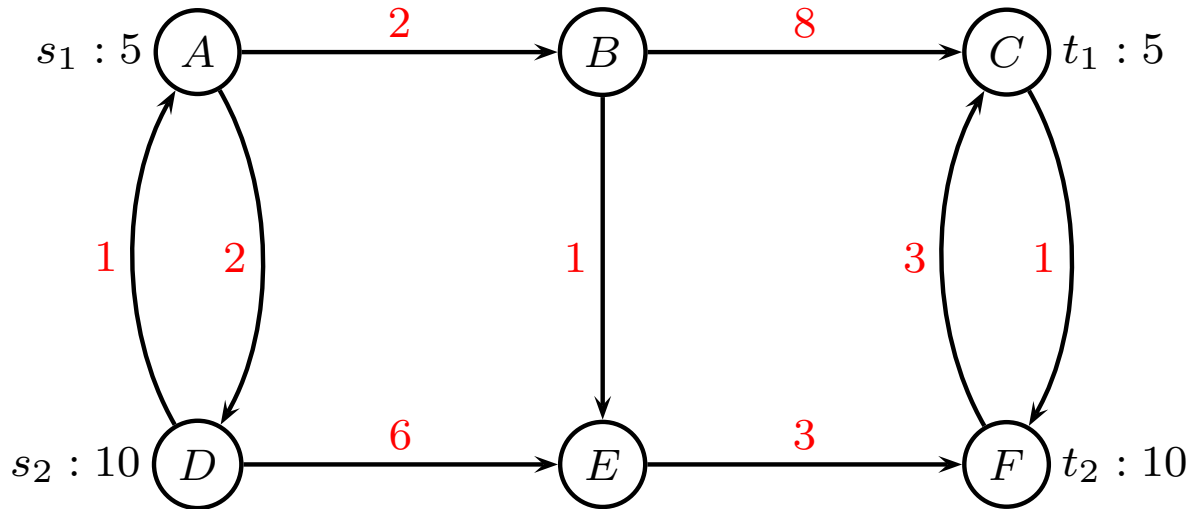
Calculate reduced costs

Reduced costs $\bar{c}_p = \sum_{(i,j) \in p} (c_{ij} - y_{ij}) - y_k^*$

path number	commodity k	path p	$\sum(c_{ij} - y_{ij})$	$\bar{c}_p$
1	1	A→B→C	10	0
2	1	A→B→E →F→C	9	-1
3	1	A→D→E →F→C	14	4
4	2	D→E→F	9	0
5	2	D→A→B→C→F	12	3
6	2	D→A→B→E→F	7	-2

Shortest path problem

Alternatively, find shortest path for each commodity



Commodity 1: $A \rightarrow B \rightarrow E \rightarrow F \rightarrow C$ cost 9

Commodity 2: $D \rightarrow A \rightarrow B \rightarrow E \rightarrow F$ cost 7

subtract $y_k^* = (10, 9)$ from path length

Second iteration

Add path 6. Restricted master problem:

minimize

$$10x_1 + 9x_4 + 7x_6$$

subject to

$$\text{AB: } +x_1 \quad +x_6 \leq 10 \quad (y_{AB})$$

$$\text{BC: } +x_1 \leq 7 \quad (y_{BC})$$

$$\text{DE: } \quad +x_4 \leq 10 \quad (y_{DE})$$

$$\text{EF: } \quad +x_4 + x_6 \leq 13 \quad (y_{EF})$$

$$\text{BE: } \quad +x_6 \leq 7 \quad (y_{BE})$$

$$\text{AD: } \leq 6 \quad (y_{AD})$$

$$\text{DA: } \quad +x_6 \leq 9 \quad (y_{DA})$$

$$\text{CF: } \leq 8 \quad (y_{CF})$$

$$\text{FC: } \leq 4 \quad (y_{FC})$$

$$\text{K1: } x_1 \geq 5 \quad (y^*_1)$$

$$\text{K2: } x_4 + x_6 \geq 10 \quad (y^*_2)$$

Solution: $x_1 = 5, x_4 = 5, x_6 = 5$

Dual: $y_1^* = 12, y_2^* = 9 \quad y_{AB} = -2$

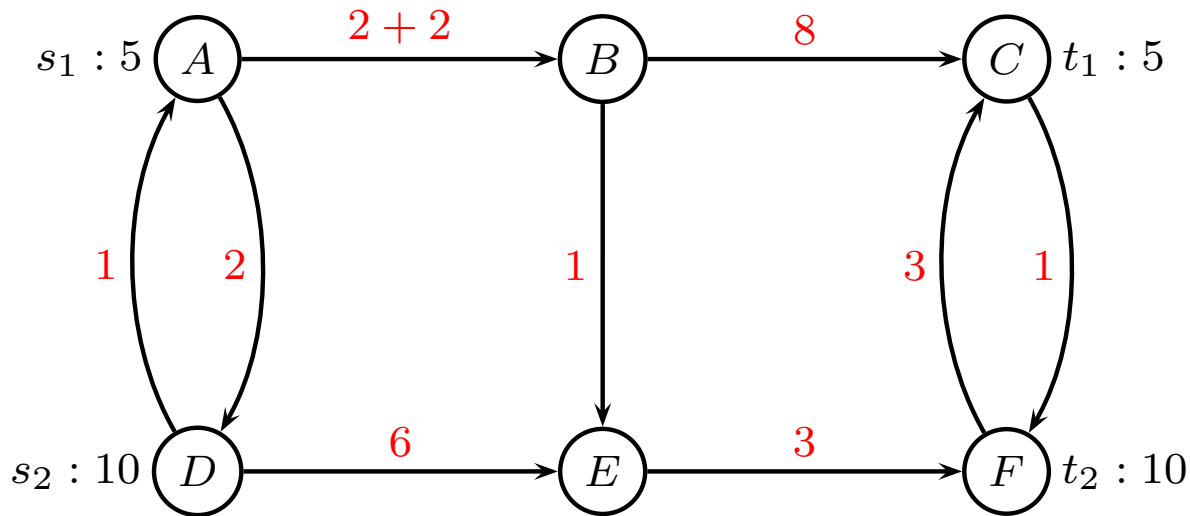
Calculate reduced costs

Reduced costs $\bar{c}_p = \sum_{(i,j) \in p} (c_{ij} - y_{ij}) - y_k^*$

path number	commodity k	path p	$\sum(c_{ij} - y_{ij})$	$\bar{c}_p$
1	1	A→B→C	12	0
2	1	A→B→E →F→C	11	-1
3	1	A→D→E →F→C	14	2
4	2	D→E→F	9	0
5	2	D→A→B→C→F	14	5
6	2	D→A→B→E→F	9	0

Shortest path problem

Alternatively, find shortest path for each commodity



Commodity 1: $A \rightarrow B \rightarrow E \rightarrow F \rightarrow C$ cost 11

Commodity 2: $D \rightarrow E \rightarrow F$ cost 9

subtract $y_k^* = (12, 9)$ from path length

Third iteration

Add path 2. Restricted master problem:

minimize

$$10x_1 + 9x_2 + 9x_4 + 7x_6$$

subject to

$$\text{AB: } +x_1 + x_2 + x_6 \leq 10 \quad (y_{AB})$$

$$\text{BC: } +x_1 \leq 7 \quad (y_{BC})$$

$$\text{DE: } +x_4 \leq 10 \quad (y_{DE})$$

$$\text{EF: } +x_2 + x_4 + x_6 \leq 13 \quad (y_{EF})$$

$$\text{BE: } +x_2 + x_6 \leq 7 \quad (y_{BE})$$

$$\text{AD: } \leq 6 \quad (y_{AD})$$

$$\text{DA: } +x_6 \leq 9 \quad (y_{DA})$$

$$\text{CF: } \leq 8 \quad (y_{CF})$$

$$\text{FC: } +x_2 \leq 4 \quad (y_{FC})$$

$$\text{K1: } x_1 + x_2 \geq 5 \quad (y^*_1)$$

$$\text{K2: } x_4 + x_6 \geq 10 \quad (y^*_2)$$

Solution: $x_1 = 3, x_2 = 2, x_4 = 5, x_6 = 5$

Dual: $y_1^* = 11, y_2^* = 9 \quad y_{AB} = -1, y_{BE} = -1$

Calculate reduced costs

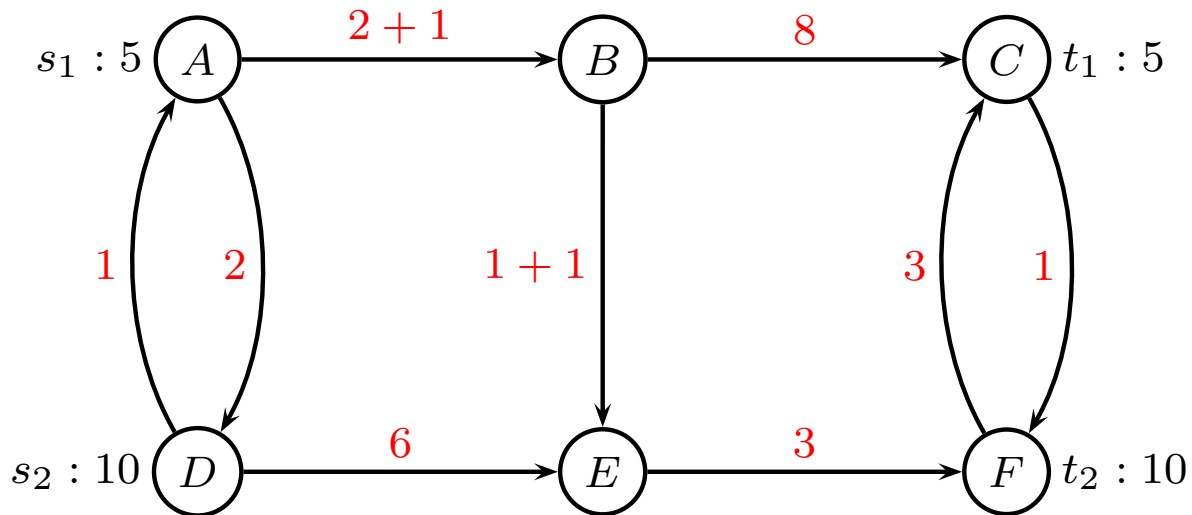
Reduced costs $\bar{c}_p = \sum_{(i,j) \in p} (c_{ij} - y_{ij}) - y_k^*$

path number	commodity k	path p	$\sum(c_{ij} - y_{ij})$	$\bar{c}_p$
1	1	A→B→C	11	0
2	1	A→B→E →F→C	11	0
3	1	A→D→E →F→C	14	3
4	2	D→E→F	9	0
5	2	D→A→B→C→F	13	4
6	2	D→A→B→E→F	9	0

No negative reduced costs, stop.

Shortest path problem

Alternatively, find shortest path for each commodity



Commodity 1: $A \rightarrow B \rightarrow C$ cost 11

Commodity 2: $D \rightarrow E \rightarrow F$ cost 9

subtract $y_k^* = (11, 9)$ from path length

No negative reduced cost, stop

Column Generation - summing up

Solution: $x_1 = 3, x_2 = 2, x_4 = 5, x_6 = 5$

Objective: $z = 10 \cdot 3 + 9 \cdot 2 + 9 \cdot 5 + 7 \cdot 5 = 128$

Generated only 4 out of 6 paths

In general:

- Interaction between master problem and subproblem
- Add all paths with negative reduced cost to master problem
- Stabilization: Dual variables may fluctuate in the beginning, add soft constraints to dual variables

Column Generation - advantages

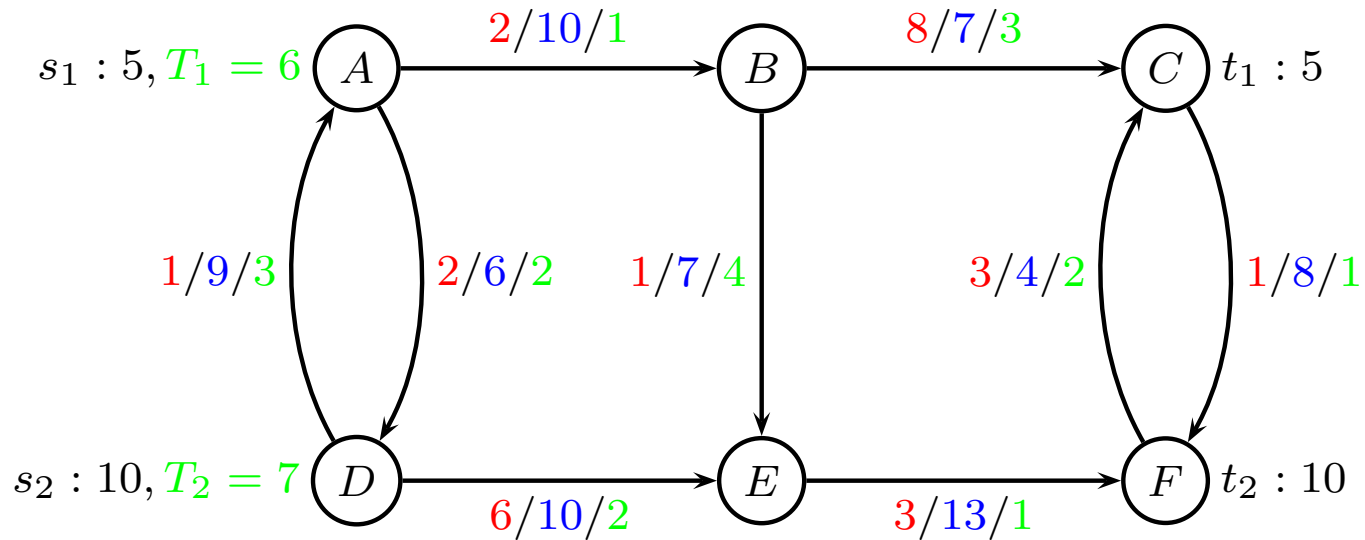
According to Ahuja, better than LP for large problems

Often easier to handle extra constraints

- min cost
- variable cost for each k
- max flow (costs ignored)
- splittable
- unsplittable
- k -splittable

Max travel time can be built into shortest-path algorithm

Time constrained MCFP



DEF is faster

DABEF is cheaper

Compact formulation ?

$$\begin{array}{ll}\min & \sum_{k \in K} \sum_{(i,j) \in E} c_{ij} x_{ij}^k \\ \text{s.t.} & \sum_{i \in V} x_{ij}^k - \sum_{i \in V} x_{ji}^k = 0 \quad k \in K, j \in V \setminus \{s_k, t_k\} \\ & \sum_{i \in V} x_{ij}^k - \sum_{i \in V} x_{ji}^k = -d_k \quad k \in K, j = s_k \\ & \sum_{i \in V} x_{ij}^k - \sum_{i \in V} x_{ji}^k = d_k \quad k \in K, j = t_k \\ & \sum_{k \in K} x_{ij}^k \leq u_{ij} \quad (i, j) \in E \\ & 0 \leq x_{ij}^k \leq d_k \quad k \in K, (i, j) \in E\end{array}$$

Difficult to handle aggregated time.

Need extra index on variables x_{ij}^{kt}

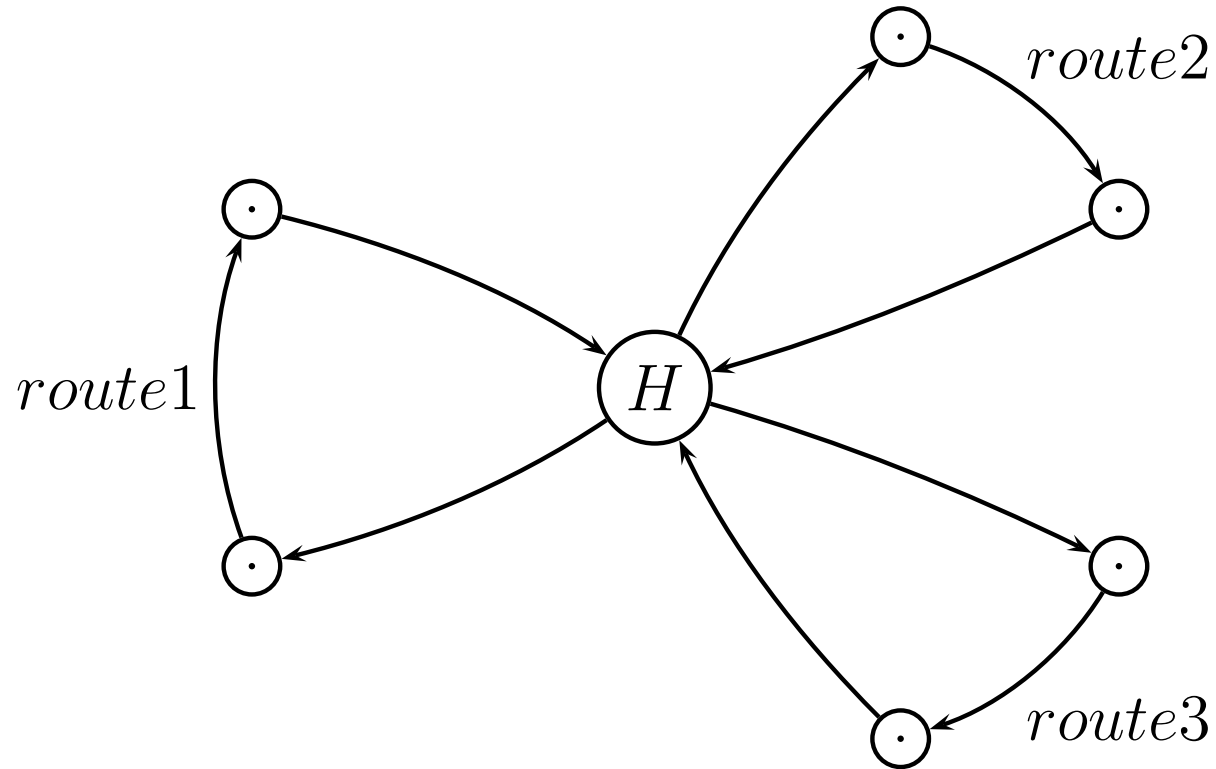
Path formulation

P_k set of possible paths from s_k to t_k **satisfying time constraint**

$$\begin{aligned} \min \quad & \sum_{k \in K} \sum_{p \in P_k} \hat{c}_p^k x_p^k \\ \text{s.t.} \quad & \sum_{k \in K} \sum_{p \in P_k} a_{p,ij}^k x_p^k \leq u_{ij} \quad (i, j) \in E \\ & \sum_{p \in P_k} x_p^k = d_k \quad k \in K \\ & 0 \leq x_p^k \leq d_k \quad k \in K, p \in P_k \end{aligned}$$

Finding paths is now time-constrained shortest path problem (NP-hard)

Modeling a hub port



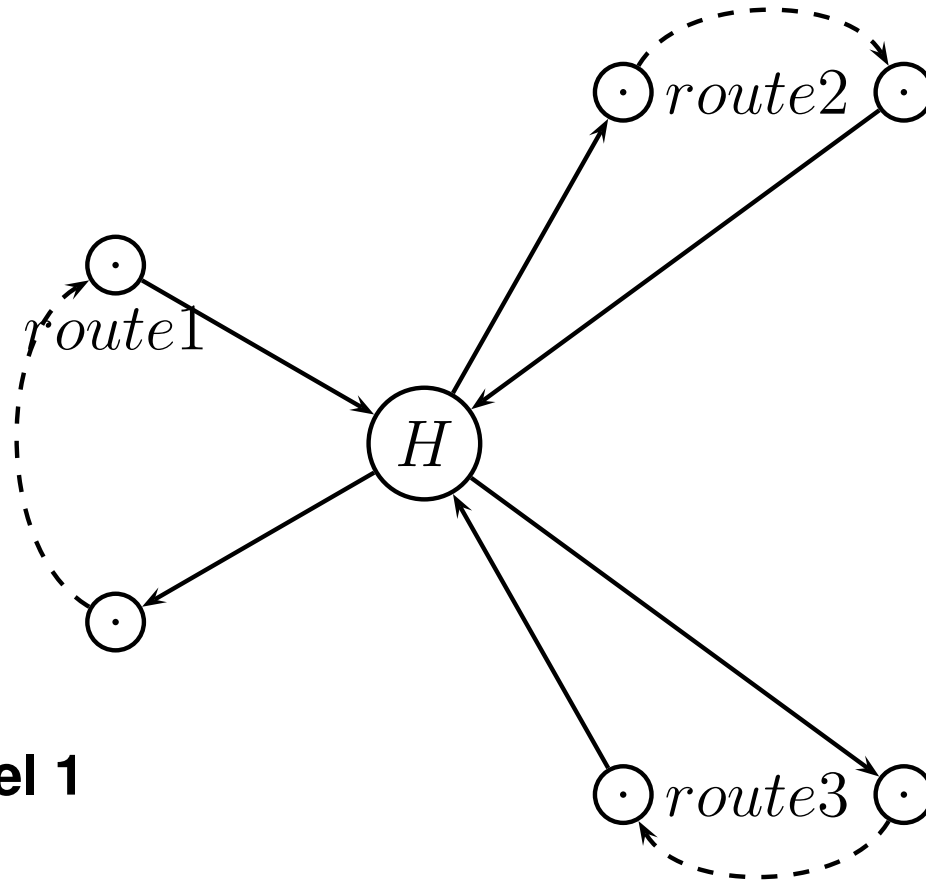
route	weekday
1	Saturday
2	Wednesday
3	Friday

Modeling a hub port



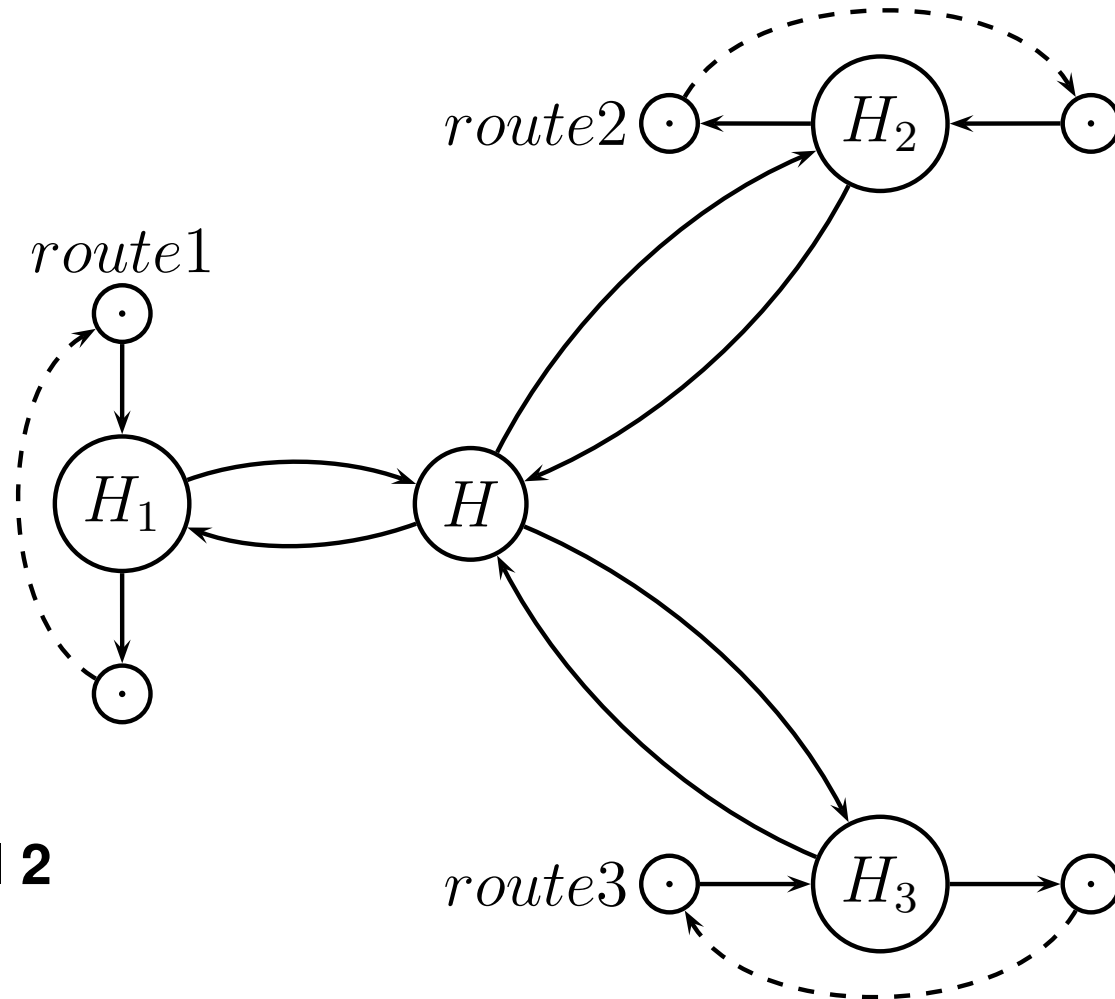
- Crane cost c (per container)
- Crane capacity u
- Transshipment time t_{ij}
- Storage cost $\bar{c}$ (per container)
- Storage capacity $\bar{u}$

Modeling a hub port



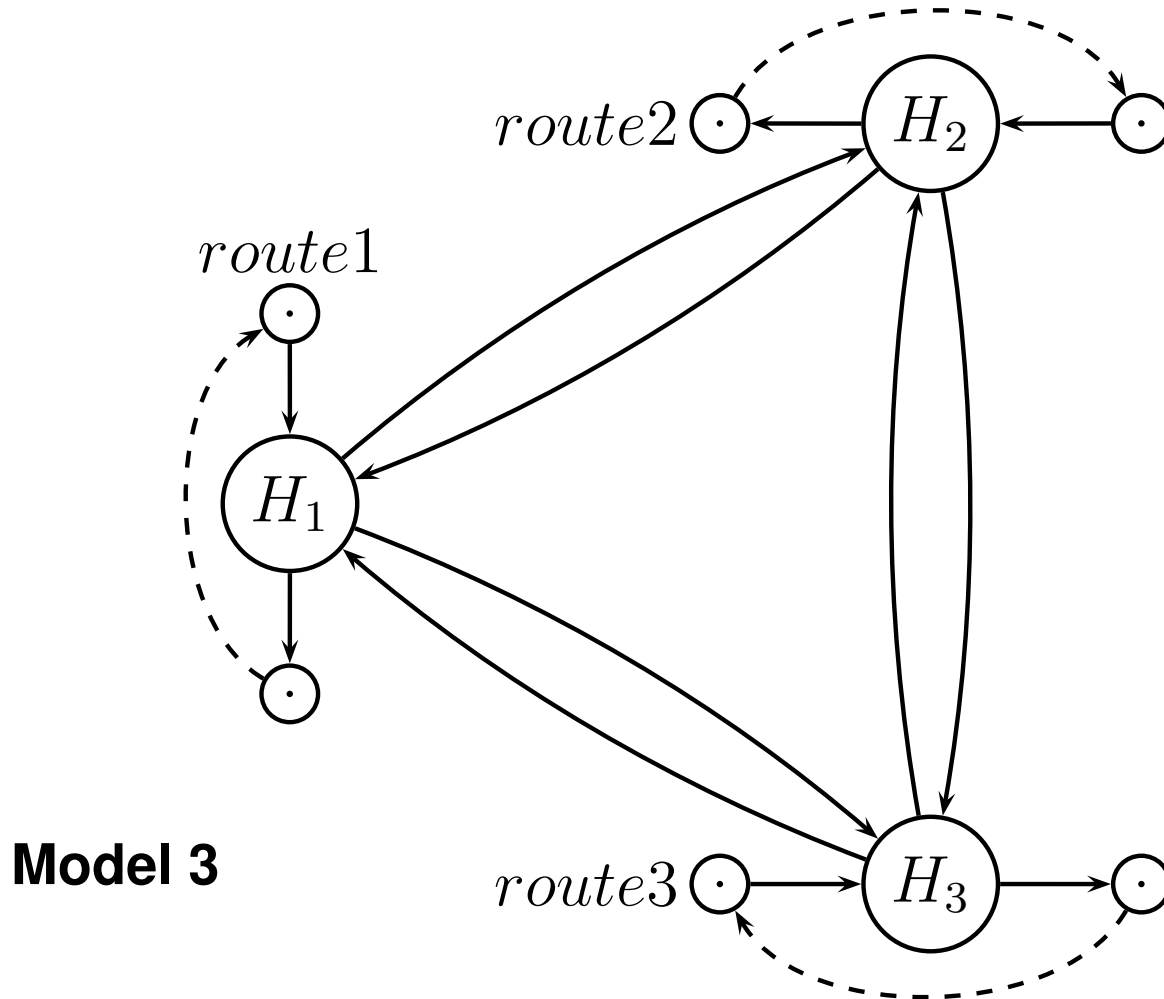
Model 1

Modeling a hub port

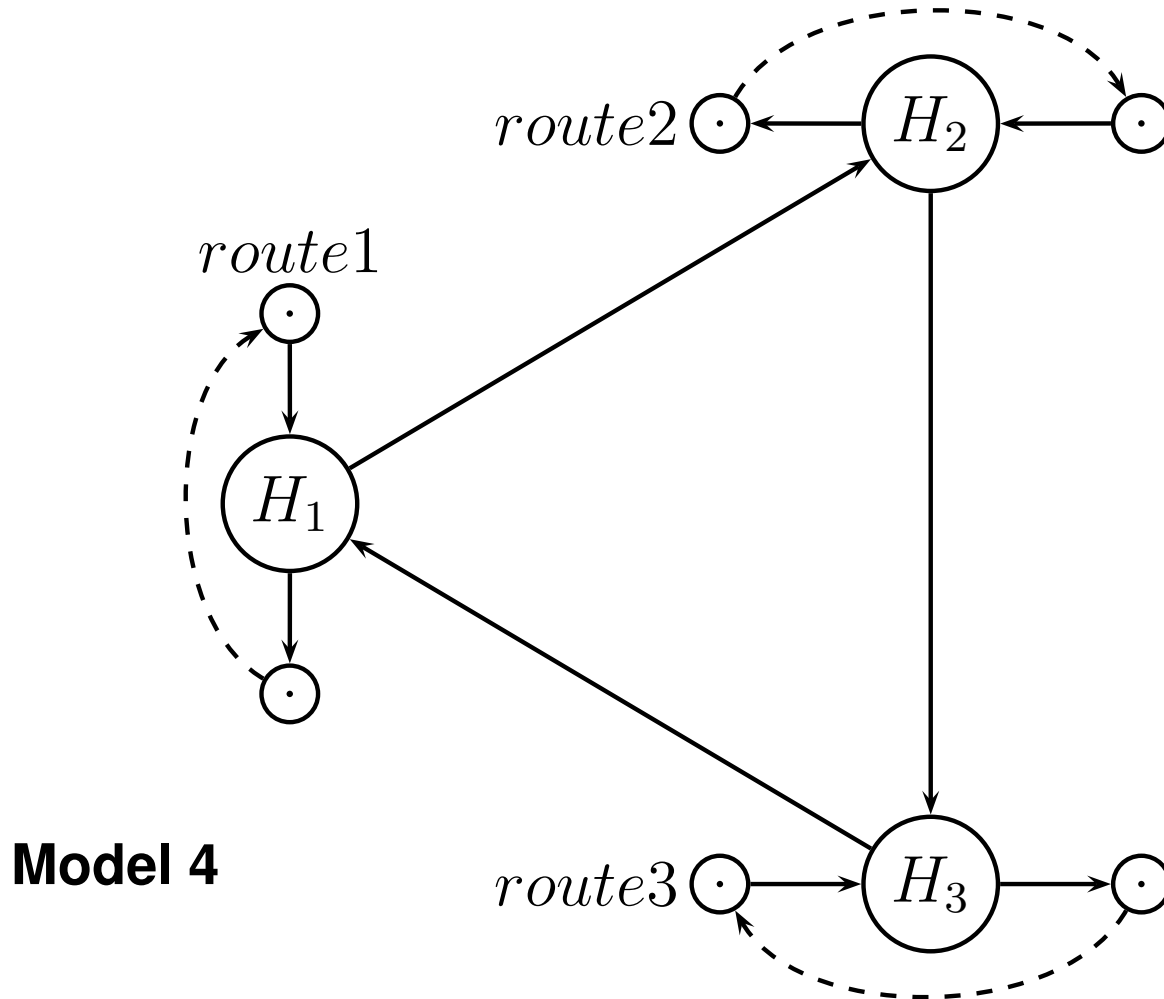


Model 2

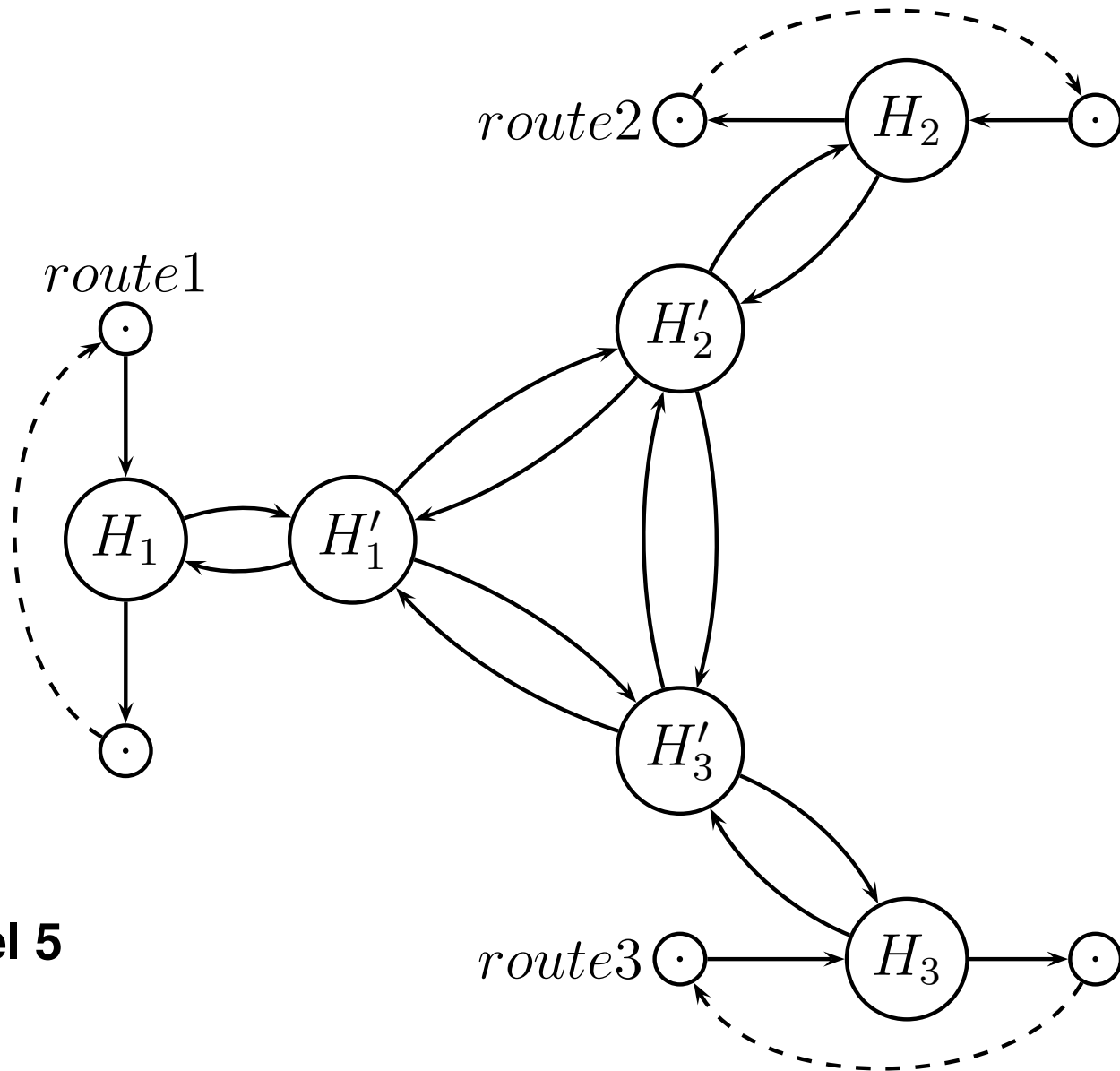
Modeling a hub port



Modeling a hub port

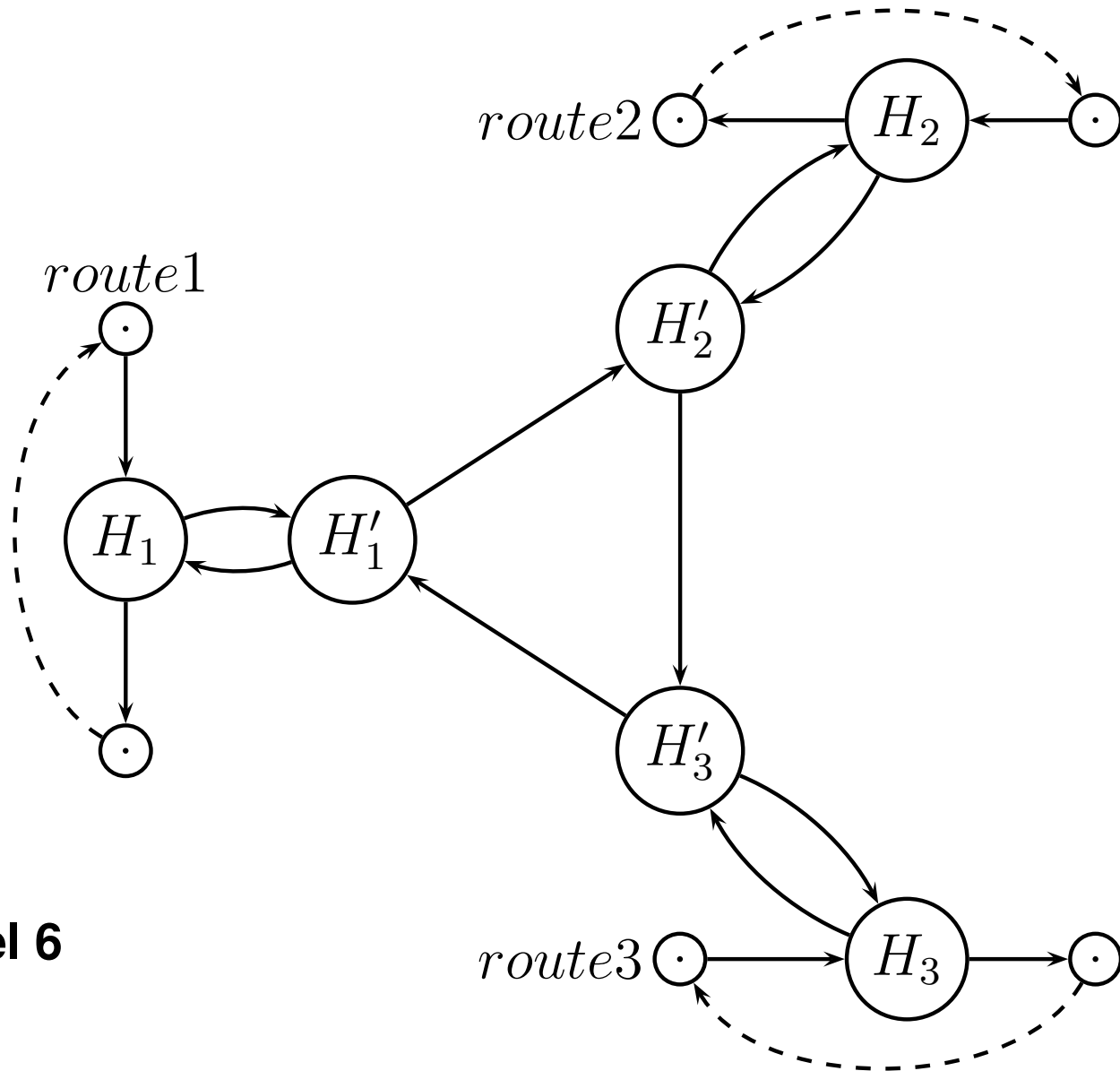


Modeling a hub port



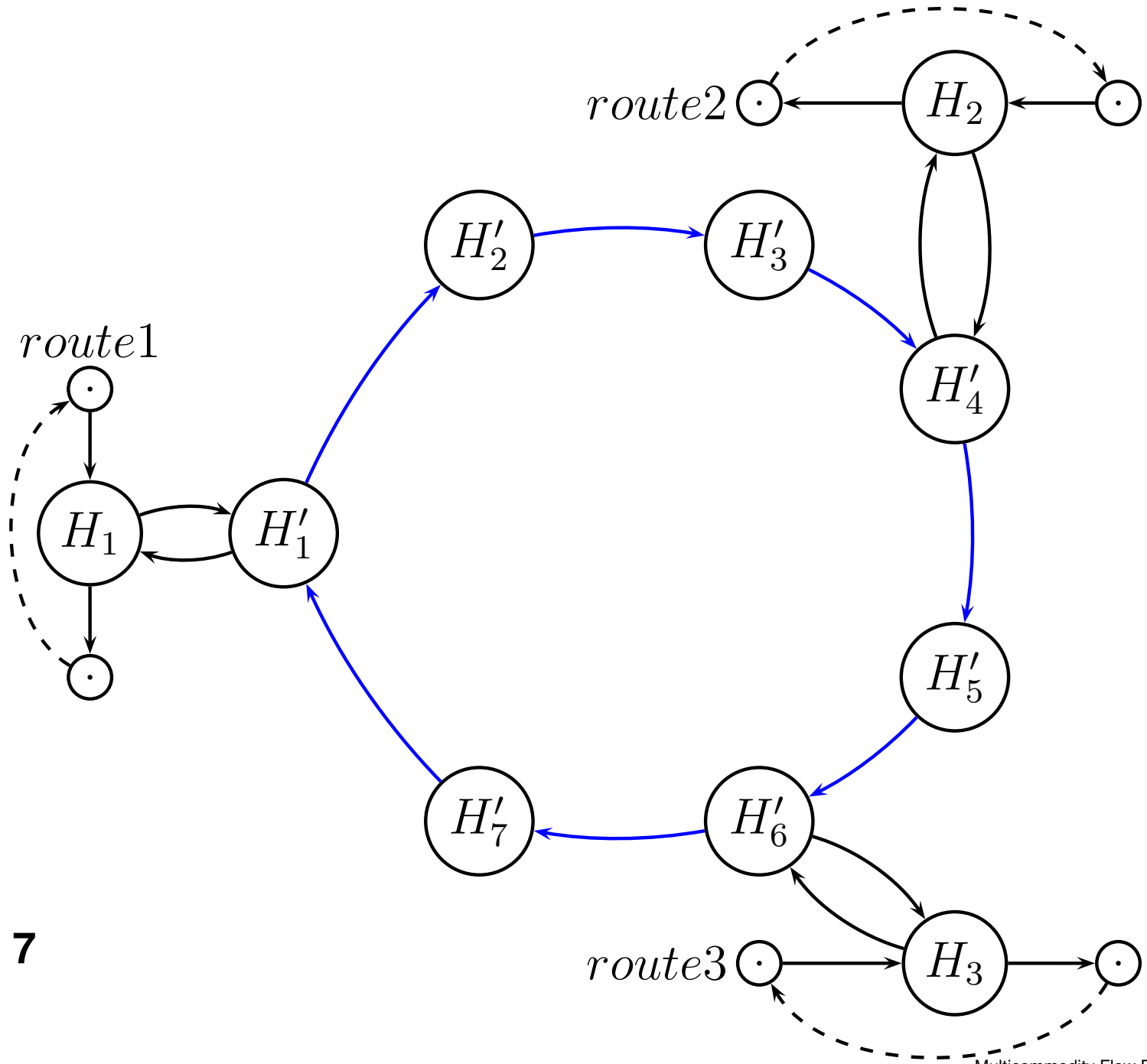
Model 5

Modeling a hub port



Model 6

Modeling a hub port



Model 7